

MAHMOUD OSAMA

AI Engineer – Client Delivery & Applied AI

Mansoura, Egypt • Open to Relocation & Remote

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PROFILE

AI engineer with freelance experience delivering AI applications end-to-end, from client requirements and technical feasibility through architecture, development, validation, deployment, and support. Builds production systems with Python, FastAPI, Django, LLM APIs, and agent workflows, including tool calling and multi-agent systems. Experienced in translating ambiguous problems into practical AI solutions, integrating models with REST APIs, testing outputs before production, and communicating technical decisions clearly to clients and non-technical stakeholders.

EXPERIENCE

AI Engineer – Freelance / Self-Employed

Jan 2022 – Present

Remote

- Delivered AI, ML, automation, and security-focused applications end-to-end, working directly with clients from requirements gathering and solution scoping through architecture, development, deployment, and ongoing support.
- Evaluated client requirements, data, systems, APIs, and technical constraints to determine feasibility, identify integration risks, and select practical solutions before committing to implementation.
- Designed AI workflows using agentic systems, tool calling, multi-agent orchestration, deterministic pipelines, and human-in-the-loop approaches, selecting architectures based on reliability, complexity, and client requirements.
- Built Python-based applications and REST APIs using FastAPI, Django, and Flask, integrating ML models and LLM-powered components for real-time inference, automation, and system workflows.
- Trained and evaluated classical ML and deep learning models (classification, CNNs) using scikit-learn and PyTorch, covering data preprocessing, feature engineering, training, and performance evaluation before deployment.
- Developed and deployed production AI systems including patient communication automation, phishing detection, security testing, and LLM-powered vulnerability analysis.
- Validated AI and ML outputs before deployment by testing accuracy, false positives, failure cases, and edge-case behaviour; debugged and optimized models and application performance before production use.
- Managed project repositories, environments, deployments, Docker containers, Git/GitHub workflows, technical documentation, and client handover and support.

SELECTED PROJECTS

WhatsApp Clinic AI Agent

GitHub

Python • NLP • WhatsApp API • Backend Systems • Client Delivery

- Scoped, built, and deployed a patient communication and scheduling system for a clinic use case, integrating real-time WhatsApp messaging with backend workflows and patient data.
- Designed the system to automate two-way patient communication and appointment-related workflows while providing physicians with visibility into appointment status and patient information.

AI Red Teaming Agent

GitHub

Python • LLMs • Multi-Agent Systems • Tool Calling • AI Evaluation

- Designed an automated agent workflow for continuous security testing, coordinating attack simulation, evaluation, and adaptive scenario generation to test system defenses.

- Applied agentic design and validation principles to evaluate security scenarios before production-style use; the project evaluation reported a 25% reduction in successful phishing attempts over three months.

Closed-Loop Intent-Based 6G Network – RL Agent

GitHub

Python • Reinforcement Learning • AI Agents • Network Automation • Closed-Loop Systems

- Built an AI-driven closed-loop network management system that translates high-level network intents into automated actions across a simulated 6G network environment.
- Developed a reinforcement learning agent to observe network conditions, evaluate state and performance metrics, and select actions that adapt network behaviour toward defined service objectives.
- Designed the system around continuous monitoring, decision-making, execution, and feedback, applying validation and evaluation to assess agent decisions before and during automated network control.
- Integrated the RL-based decision layer with network simulation and SDN components to demonstrate how AI agents can support autonomous network management and optimization.

EDUCATION

Delta University for Science and Technology

Jun 2026

Mansoura, Egypt

BSc in Artificial Intelligence & Cybersecurity • GPA: 3.2 / 4.0

- Certifications: NVIDIA Deep Learning Institute (DLI) Summer Training (ITI) • Building LLM Applications with Prompt Engineering • Building Agentic AI (Andrew Ng)

SKILLS

AI & LLMs

LangChain, OpenAI API, LLM Applications, Agent Workflows, Tool Calling, Multi-Agent Systems, Prompt Engineering, Model Selection, Human-in-the-Loop Workflows

Machine Learning

PyTorch, scikit-learn, CNNs, Classification Models, Reinforcement Learning (DQN Agents), Model Training & Evaluation

AI Engineering & Validation

AI Output Validation, Failure-Mode Testing, Edge-Case Testing, Accuracy Evaluation, False-Positive Reduction, ML Pipelines

Client Delivery

Requirements Discovery, Technical Feasibility, Solution Scoping, Architecture, Documentation, Client Onboarding, Technical Communication

Backend & APIs

Python, FastAPI, Django, Flask, REST APIs, Real-Time Inference

Cloud & Delivery

Google Cloud Platform, Docker, Git, GitHub, CI/CD, Deployment, Environment Management

Data & Systems

Data Processing, Feature Engineering, Model Integration, System Architecture, Security Automation